

Christopher (Christy) Bruce

christy.108@outlook.com | +44 7305 223429 | www.linkedin.com/in/christopher-bruce-943202224

EDUCATION

Loughborough University <i>MSc. Financial Mathematics</i> Distinction (80.5% , Course Rank: 2nd , Scholarship) <i>BSc. Mathematics Major (Natural Sciences)</i> First-class honours.	Loughborough, UK <i>Sep 2023 – Oct 2024</i> <i>Sep 2019 – Jun 2022</i>
---	---

Relevant Modules: Statistics for Large Data (Data Science, XGBoost, Random Forests) (83%), Timeseries Econometrics (95%), Economic Data Analysis (90%), Probability Theory (79%), Derivative Pricing (73%), Financial Economics (76%), Computational Finance (83%),

Masters Thesis Paper for Publication (Complex Systems & Networks) (Scored 93%)

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Statsmodels, Scikit-Learn), R, SQL.
Quantitative/Statistical: Econometrics, Time-Series Analysis, Machine Learning (XGBoost, Random Forests).
Finance/Research: ESG Data Analysis, Factor Modelling, Portfolio Construction, Derivative Pricing.

PROFESSIONAL EXPERIENCE

Imperial College London Quantitative Investing - Research Consultant	London <i>Feb 2026 - Present</i>
--	--

- Partnered with Imperial’s Data, Systems, and Finance Labs, producing equity portfolios that is currently beating the benchmark by 2.8% annually in the US market.
- Responsible for factor and signal research from NLP derived signals, revealing key drivers of excess returns.
- Engineered regression models to link behavioural signals to predict equity accounting performance.
- Present findings commonly to senior management simplifying complex topics in simple relevant terms.
- Built automated pipelines in Pandas/SQL to synchronize disparate datasets from multiple sources (e.g. WRDS).
- Collaborating with Nikkei Exchange Japanese analysts to build a sustainability-linked tradable index.
- Built a comprehensive strategy performance suite (e.g. VaR, Rolling Sharpe, Fama-French).

All Options Quantitative Trader Programme	Amsterdam <i>Sept 2025 - Oct 2025</i>
---	---

- Selected with a 0.5% acceptance rate to the programme, involving an intensive course in **option theory**, along with fast-paced market-making games and hands-on shadowing of options traders and quantitative researchers.
- Enhanced my ability to tackle complex problems in a fast-changing environment.

Introspect Capital Quantitative Analyst Intern	Dubai - London <i>May 2024 - Oct 2024</i>
--	---

- Algorithmic Modelling & Risk Analytics:** Developed an unsupervised learning (Network Clustering) framework to identify hidden market structures, contributing to more accurate risk assessment.
- Data Manipulation:** Designed SQL and Python pipelines to synchronise and clean factor data for 500+ stocks, ensuring data integrity for back-testing and validating using Pandas and Numpy.
- Machine Learning:** leveraged XGBoost for predictive modeling of financial time-series, focusing on minimizing model error and understanding feature importance for stakeholder reporting.
- Communication/Team:** Communicated complex technical findings through data visualization and comprehensive research reports for senior management; collaborated with cross-functional teams.

RESEARCH PROJECTS & COMPETITIONS

Masters Thesis for Conference Publication (Complex Systems & Networks) (Scored 93%)

Loughborough

Researcher

Jun 2023 – Present

- Built a Python-based algorithm to dynamically track changes in correlation structure in equity markets.
- Analysed ~2,500 stocks over two years to identify how systemic shocks (e.g. Covid-19) altered clustering, risk concentration, and market leadership.
- Used clustering and network metrics to detect regime shifts and periods of elevated systemic risk, supporting more informed **risk-adjusted investment decisions**.
- Presented my findings to the NetSci conference with an audience of over 50 professionals, explaining complex topics into clear, actionable insights.

Custom Machine Learning Model from Scratch (Python)

Amsterdam

Researcher

April 2025 – Jan 2026

- Built my own invented machine learning model from scratch to predict time series data that calculated the expected value using conditional probabilities of binary past pattern averages.
- Performance in trading had a 2.7 in-trade Sharpe ratio on 6 years of out-of-sample daily data on the US market index, based on custom-built back testing engines.
- Implement dynamic trade size based on predicted volatility using GARCH and predicted confidence.
- Presented findings to Amsterdam Investment Club, winning an award for exceptional participation.

IMC Prosperity, Algorithmic Trading Challenge

London

Co-Researcher

April 2025 – May 2025

- Achieved top 3% of 12000+ teams in a 15-day trading challenge of 5 rounds with 2 collaborators.
- Implemented derivative pricing via Black-Scholes to make a trading algorithm based on relative value.
- Created arbitrage strategies between assets, managing risk, e.g. by dynamically adjusting position size.

Prediction Market Systematic Model (Event-Driven) (Python)

Amsterdam

Co-Researcher

Oct 2025 – Jan 2026

- Engineered an event-driven statistical model to identify mainsprings in event-contract markets by deriving implied probabilities and comparing them to market consensus.
- Developed automated API-based pipelines for real-time data ingestion.
- Executed a 40% absolute return over a 2-month horizon (75 trades), maintaining top 5% performance rank in 2025 on Polymarket.

LEADERSHIP & EXTRACURRICULAR

Amsterdam Sahaj Yoga Meditation Course

Amsterdam

Co-Organiser

Oct 2025 – Present

- Lead various meditation workshops weekly, giving presentations on topics related to meditation and balance
- Guiding participants one-to-one in their path to self-mastery, balance and a clear mind through meditation.

International Finance Conference

French Alps

Co-Organiser

Jan 2025 – Jun 2024

- *Team Leadership/Communication:* Initiated and co-organised a Quant Finance research conference, presenting 4 personal research projects and coordinating 10 attendees.